Experiment 03 – Lab 5

Compare ChatGPT's Code with Expected Output

Aim

To generate a simple program using ChatGPT, run or inspect the code, and compare its actual output with the expected output to identify whether the result is correct.

Task

Use ChatGPT to generate a Python program that calculates the total and average of three marks. Compare the expected output with the output produced by the code.

Prompt given to ChatGPT

Write a simple Python program that accepts marks in three subjects, calculates the total and average, and displays the results clearly.

Code generated by ChatGPT

```
marks1 = float(input('Enter marks for Subject 1: '))
marks2 = float(input('Enter marks for Subject 2: '))
marks3 = float(input('Enter marks for Subject 3: '))

total = marks1 + marks2 + marks3
average = total / 3

print('Total marks:', total)
print('Average marks:', average)
```

Test Input

Subject 1 = 80, Subject 2 = 75, Subject 3 = 85

Expected Output

```
Enter marks for Subject 1: 80
Enter marks for Subject 2: 75
Enter marks for Subject 3: 85
Total marks: 240.0
Average marks: 80.0
```

Actual Output

For the test inputs 80, 75, and 85, the program is expected to display:
Total marks: 240.0
Average marks: 80.0

Comparison: Expected vs Actual

Check	Expected	Actual / Observed	Status
Total	240.0	240.0	Match
Average	80.0	80.0	Match
Calculation	Add three marks and divide by 3	Same formula used in code	Correct
Formatting	Clear labels for total and average	Labels are displayed	Match

Analysis

The program uses float values so that marks can include decimal values. It correctly adds the three subject marks and divides the total by 3 to calculate the average. For the sample inputs 80, 75, and 85, the total is 240 and the average is 80. The expected and calculated results match.

Conclusion

ChatGPT generated code that follows the requested logic. By checking the expected output against the calculated output for the given test input, the program's basic correctness can be verified. More test cases, including decimal marks and invalid input, would be useful for more complete testing.

Result

Successfully compared ChatGPT-generated Python code with the expected output for a sample input and found that the results match.

Note: The output shown is calculated from the code for the stated test inputs; it is not a screenshot of a live execution.